The empirical recurrence for OEIS A253858: recovering a conjecture that is not written down

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Abstract

OEIS A253858 records "Empirical recurrence of order 70", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 234$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 70, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 70 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A253858 is "Number of (n+2)X(3+2) 0..1 arrays with every 3X3 subblock diagonal maximum plus antidiagonal minimum nondecreasing horizontally and vertically.". It has offset 1 and begins

$$19456, 291392, 4186496, 54933632, 700079360, 9090451328, 118920758784, 1541740409984, \ \ldots$$

The entry states:

> Empirical recurrence of order 70 (see link above)

As of the "Last modified" line on the live entry (Jul 23 2025 14:18:18, revision 6) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 1632$ vertices, $a(n) = \iota^\top M^{j(n)}\tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 234$ of the 1632, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' =$ 234.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 468$ exact terms of the model returns order 70 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{70} c_i\, a(n-i),$$

c_1	24	c_{19}	−7102960106471424	c_{37}	−20584556850832729310232576	c_{55}	−2
c_2	−154	c_{20}	33724625514921984	c_{38}	92721684390626033859035136	c_{56}	25
c_3	−52	c_{21}	−132659588490067968	c_{39}	−224366889187965148675964928	c_{57}	−
c_4	4375	c_{22}	334727066375159808	c_{40}	317013932834648785380442112	c_{58}	−7
c_5	−41308	c_{23}	−145267672756518912	c_{41}	190680039545635585872887808	c_{59}	15
c_6	540348	c_{24}	−4483595198714085376	c_{42}	−2120941065796235748813307904	c_{60}	−15
c_7	−6906304	c_{25}	30372006540997558272	c_{43}	4713578472439674057015689216	c_{61}	52
c_8	42404864	c_{26}	−142610199237183406080	c_{44}	−5207301178024048626289868800	c_{62}	97
c_9	−54831104	c_{27}	551822030157689913344	c_{45}	−159203148988624388457758720	c_{63}	−21
c_{10}	−655667200	c_{28}	−1529706504928440090624	c_{46}	13866170296943394094336442368	c_{64}	20
c_{11}	9493315584	c_{29}	3234853863650702131200	c_{47}	−23691474165851130802112299008	c_{65}	−9
c_{12}	−83166920704	c_{30}	−3800955127348382924800	c_{48}	4794296683693000111532539904	c_{66}	−4
c_{13}	491754618880	c_{31}	−10162884636910951596032	c_{49}	40025317916968285393504960512	c_{67}	36
c_{14}	−2419059195904	c_{32}	37747080770319095431168	c_{50}	−75749382840401964033011351552	c_{68}	−3
c_{15}	8384201621504	c_{33}	35201680055518682415104	c_{51}	69742429962720772906406641664	c_{69}	15
c_{16}	−12378541391872	c_{34}	−420485576466261153415168	c_{52}	28995953557195360405261647872	c_{70}	−3
c_{17}	−107492491132928	c_{35}	1435478999737985000275968	c_{53}	−379179281090673536989298425856		
c_{18}	1253722072023040	c_{36}	−505162678520702564302848	c_{54}	1183413026906800670174793957376		

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 70, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $70 + 4$ terms removed returns order 70 as well, so $\deg q_{\min} = 70 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. □

4 Verification

Three checks.

First, the walk model reproduces all 14 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $70 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 70 that this sequence satisfies — and that family turns out to have exactly one member.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A253858.

[2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.

[3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)

[4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.